

Literature Status Note: The 3D Heisenberg L_1 Distortion Conjecture

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Status

Status: literature_already_answered. The L_1 -distortion conjecture in Lafforgue–Naor, *Vertical versus horizontal Poincaré inequalities on the Heisenberg group*, arXiv:1212.2107, is answered negatively by Naor–Young, *Foliated corona decompositions*, arXiv:2004.12522.

Source Question

In Section 4, Lafforgue–Naor ask whether the endpoint $p = 1$ vertical-versus-horizontal inequality holds for smooth compactly supported real-valued functions on the real 3-dimensional Heisenberg group. They reformulate this as an isoperimetric question and state Conjecture 4.4:

$$c_1(B_n, d_W) \asymp \sqrt{\log n} \quad (n \geq 2),$$

where B_n is the radius n word ball in the discrete 3-dimensional Heisenberg group.

Supporting Answer

The supporting paper is Assaf Naor and Robert Young, *Foliated corona decompositions*, arXiv:2004.12522. Its introduction explicitly says that the asymptotic evaluation of $c_{\ell_1}(\mathcal{B}_n)$ had been asked in the Lafforgue–Naor line of references and that most of those references conjectured the $\sqrt{\log n}$ behavior.

The decisive statement is Theorem 1.6:

$$c_{\ell_1}(\mathcal{B}_n) \asymp \sqrt[4]{\log n} \quad \text{for every integer } n \geq 2.$$

Here $\mathcal{B}_n$ is the word ball in the same discrete 3-dimensional Heisenberg group.

Conclusion

Naor–Young’s two-sided estimate gives the exact exponent $1/4$. Therefore the Lafforgue–Naor conjectural conclusion

$$c_1(B_n, d_W) \asymp \sqrt{\log n}$$

is false for the 3-dimensional Heisenberg group. Consequently the endpoint Question 4.1, the isoperimetric Question 4.2, and the coarse total vertical perimeter Conjecture 4.3 cannot all hold in a form that implies Conjecture 4.4 as stated.

Scope

This is a literature answer rather than a new proof from the run. It concerns the 3-dimensional Heisenberg group. Naor–Young also stress that the 5-dimensional Heisenberg group has L_1 -distortion of order $\sqrt{\log n}$, so the dimension is essential.

References

- [1] V. Lafforgue and A. Naor, *Vertical versus horizontal Poincaré inequalities on the Heisenberg group*, Israel J. Math. 203 (2014), 309–339; arXiv:1212.2107.
- [2] A. Naor and R. Young, *Foliated corona decompositions*, arXiv:2004.12522.